

Generation Interconnection System Impact Study Report

For Queue Project Q18-0920

Ibm Bailey 230 kV
November 24, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0920. The Interconnection Customer proposes a 220.8 MW solar photovoltaic facility interconnecting at the Ibm Bailey 230 kV substation in the Illinois area. The request was received on January 13, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 282 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,868,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The DIXON LANDING / TROY ENERGY LLC 230 kV is overloaded to 212.6% of its 675 MVA emergency rating for the loss of the TROY ENERGY LLC / RICHLAND 500 kV. The project contributes 8.6% DFAX (18.9 MW) to this overload.

HIGHGATE / TREGO SUBSTATION 230 kV line: 129.2% loading (777 MVA) following the outage of the TREGO / TROY ENERGY LLC 500 kV line; DFAX = 10.9%.

For the loss of TROY ENERGY LLC / TREGO 500 kV, the TREGO 500/1 kV Transformer loads to 125.1% of its 977 MVA rating. Project DFAX: 12.9%.

TREGO 230/1kV Transformer Bank: 122.2% loading (1,000 MVA) following the outage of the TREGO / TROY ENERGY LLC 500 kV; DFAX = 12.9%.

JOSO / EUFAULA 230 kV: 119.1% loading (783 MVA) following the outage of the BEAVER / EUFAULA 500 kV; DFAX = 8.1%.

The IMPERIL / GRUVER 230 kV is overloaded to 118.2% of its 574 MVA emergency rating for the loss of the IMPERIAL / EUFAULA 500 kV. The project contributes 6.6% DFAX (14.5 MW) to this overload.

For the loss of EUFAULA / BEAVER 500 kV L, the JOSO / ONANCOCK DIESEL GENERATION FACILITY 230 kV loads to 113.9% of its 2,010 MVA rating. Project DFAX: 14.3%.

TROY ENERGY LLC / HIGHGATE 230 kV: 112.5% loading (1,318 MVA) following the outage of the TREGO / TROY ENERGY LLC SUBSTATION 500 kV line; DFAX = 12.3%.

The GALION GENERATING STA / MYRTLE POINT 230 kV is overloaded to 112.3% of its 1,298 MVA emergency rating for the loss of the BEAVER / TREGO 500 kV. The project contributes 7.4% DFAX (16.3 MW) to this overload.

The CORRIDOR / LARSON ST 230 kV is overloaded to 112.3% of its 1,093 MVA emergency rating for the loss of the BEAVER / TREGO 500 kV. The project contributes 9.9% DFAX (21.8 MW) to this overload.

CORRIDOR / LOOKINGGLASS 230 kV L: 111.5% loading (1,127 MVA) following the outage of the TREGO / BEAVER 500 kV; DFAX = 9.9%.

TRUMAN / LOOKINGGLASS 230 kV: 110.8% loading (1,162 MVA) following the outage of the TREGO / BEAVER 500 kV; DFAX = 9.9%.

The WALDO SUBSTATION / TRUMAN 230 kV is overloaded to 110.2% of its 1,191 MVA emergency rating for the loss of the TREGO / BEAVER 500 kV. The project contributes 9.9% DFAX (21.8 MW) to this overload.

For the loss of EAST METHUEN 500/230 kV Auto, the ADIRONDACK SUBSTATION 500 kV/230 kV Auto loads to 109.0% of its 1,292 MVA rating. Project DFAX: 5.1%.

For the loss of TREGO / BEAVER 500 kV, the ADAMS / PICKFORD 230 kV loads to 108.1% of its 1,282 MVA rating. Project DFAX: 8.1%.

CHAMBERLIN / IMPERIAL 230 kV: 108.1% loading (1,299 MVA) following the outage of the HYLEBOS / RAVENNA 500 kV; DFAX = 10.7%.

For the loss of TREGO / BEAVER 500 kV Line, the WALDO 500/230kV Autotransformer loads to 108.0% of its 1,310 MVA rating. Project DFAX: 9.8%.

The BARAGA VLG / MADELINE 230 kV is overloaded to 107.9% of its 842 MVA emergency rating for the loss of the GRAY'S FLAT / EAST METHUEN 230 kV. The project contributes 6.5% DFAX (14.4 MW) to this overload.

The ADAMS / ONANCOCK DIESEL GENERATION FACILITY 230 kV is overloaded to 107.6% of its 2,527 MVA emergency rating for the loss of the BEAVER / TREGO 500 kV. The project contributes 15.9% DFAX (35.1 MW) to this overload.

For the loss of BEAVER / TREGO 500 kV Line, the PICKFORD / HERALD-TIN CITIES 230 kV loads to 107.3% of its 1,661 MVA rating. Project DFAX: 9.7%.

The MASON ST / NEW DEPARTURE 230 kV is overloaded to 107.1% of its 1,300 MVA emergency rating for the loss of the TREGO / BEAVER 500 kV. The project contributes 7.8% DFAX (17.3 MW) to this overload.

TREGO / BEAVER 500 kV: 106.9% loading (5,059 MVA) following the outage of the ONANCOCK DIESEL GENERATION FACILITY / ADAMS 230 kV; DFAX = 30.9%.

For the loss of E METHUEN 500-230 kV Auto, the CARLEY ROAD / JAMES DE YOUNG 230 kV loads to 106.1% of its 991 MVA rating. Project DFAX: 7.1%.

For the loss of BEAVER / TREGO 500 kV, the OMNI-HACKBERRY / NORWICH 500 kV loads to 106.0% of its 3,369 MVA rating. Project DFAX: 18.8%.

For the loss of TREGO / BEAVER 500 kV Line, the WEEDPATCH / CANAL ROAD 230 kV loads to 105.5% of its 1,187 MVA rating. Project DFAX: 7.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DIXON LANDING / TROY ENERGY LLC 230 kV	Rebuild line with higher-capacity conductor	\$14,435,000	2% (\$247,000)
HIGHGATE / TREGO SUBSTATION 230 kV line	Rebuild line with higher-capacity conductor	\$49,984,000	2% (\$858,000)
TREGO 500/1 kV Autotransformer	Replace transformer with higher-rated unit	\$34,182,000	2% (\$585,000)
TREGO 230/1kV Autotransformer	Replace transformer with higher-rated unit	\$11,161,000	2% (\$191,000)
JOSO / EUFAULA 230 kV L	Rebuild line with higher-capacity conductor	\$7,969,000	2% (\$135,000)
IMPERIL / GRUVER 230 kV	Rebuild line with higher-capacity conductor	\$13,881,000	2% (\$235,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
JOSO / ONANCOCK DIESEL GENERATION FACILITY 230 kV	Rebuild line with higher-capacity conductor	\$15,834,000	2% (\$268,000)
TROY ENERGY LLC / HIGHGATE 230 kV	Rebuild line with higher-capacity conductor	\$34,256,000	2% (\$586,000)
MYRTLE PT / GALION GENERATING STA 230 kV	Rebuild line with higher-capacity conductor	\$24,472,000	2% (\$415,000)
CORRIDOR / LARSON ST SUBSTATION 230 kV line	Rebuild line with higher-capacity conductor	\$12,975,000	2% (\$220,000)
CORRIDOR / LOOKINGGLASS 230 kV L	Rebuild line with higher-capacity conductor	\$21,373,000	2% (\$363,000)
TRUMAN / LOOKINGGLASS 230 kV	Rebuild line with higher-capacity conductor	\$18,026,000	2% (\$306,000)
WALDO SUBSTATION / TRUMAN 230 kV	Rebuild line with higher-capacity conductor	\$18,591,000	2% (\$316,000)
ADIRONDCK 500 kV/230 kV Auto	Replace transformer with higher-rated unit	\$35,452,000	2% (\$598,000)
ADAMS / PICKFORD 230 kV	Reconductor line	\$6,778,000	2% (\$115,000)
IMPERIAL / CHAMBERLIN 230 kV	Reconductor line	\$4,040,000	2% (\$68,000)
WALDO 500/230kV Autotransformer	Replace transformer with higher-rated unit	\$30,252,000	2% (\$513,000)
BARAGA VLG / MADELINE 230 kV	Reconductor line	\$7,838,000	2% (\$132,000)
ONANCOCK DIESEL GENERATION FACILITY / ADAMS 230 kV L	Reconductor line	\$20,767,000	2% (\$352,000)
PICKFORD / HERALD-TIN CITIES 230 kV	Reconductor line	\$18,519,000	2% (\$314,000)
NEW DEPARTURE / MASON ST 230 kV	Reconductor line	\$22,645,000	2% (\$384,000)
TREGO / BEAVER 500 kV	Reconductor line	\$99,619,000	2% (\$1,687,000)
CARLY ROAD / JAMES DE YOUNG 230 kV	Reconductor line	\$7,641,000	2% (\$129,000)
OMNI-HACKBERRY / NORWICH 500 kV L	Reconductor line	\$43,669,000	2% (\$737,000)
CANAL RD / WEEDPATCH 230 kV	Reconductor line	\$6,779,000	2% (\$114,000)
	Total allocated to project		\$9,868,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.